Code No: 135CU **R16**

JAWAHARLAL NEHRU TECHNOLOGICAL UNIVERSITY HYDERABAD
B. Tech III Year I Semester Examinations, January/February - 2023
NON-CONVENTIONAL POWER GENERATION
(Common to CE, ME, CSE)

Time: 3 Hours **Max. Marks: 75**

Note: i) Question paper consists of Part A, Part B.
ii) Part A is compulsory, which carries 25 marks. In Part A, Answer all questions.
iii) In Part B, Answer any one question from each unit. Each question carries 10 marks and may have a, b as sub questions.

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PART – A

(25 Marks)

1.a)	Explain solar pond. Write its applications	[2]
b)	What is meant by solar spectrum? Is solar spectrum continuous?	[3]
c)	Draw the equivalent circuit of PV system and explain briefly	[2]
d)	Explain how photovoltaic cell and photovoltaic module are related.	[3]
e)	What is the efficiency of a wind turbine?	[2]
f)	List out the advantages and disadvantages of wind turbines.	[3]
g)	What is the difference between biomass and biogas?	[2]
h)	Define gasifiers and digesters.	[3]
i)	What are the main components of a fuel cell system? Show by means of a schematic diagram?	[2]
j)	Briefly define the principle of tidal energy.	[3]

PART – B

(50 Marks)

2.a) Explain, with sketches, the various types of solar radiation measurement instruments.
b) Explain the working of concentrated collector with the help of a neat sketch. [5+5]

OR

3.a) Derive an expression for total radiation on an inclined surface. Show that a horizontal surface receives no ground reflected radiation.
b) Discuss the following terms: Diffuse radiation, Reflected radiation and Total radiation. [5+5]

4.a) Draw the I-V characteristics of PV cell and describe the system configuration for maximum power extraction from PV system.
b) Describe the principle of solar photovoltaic conversion. [5+5]

OR

5.a) A solar photovoltaic panel has an open circuit voltage of 43.6V and short circuit current of 5.45A. The maximum power that can be drawn from the panel is 175W. The maximum power occurs at a voltage of 81% of open circuit voltage. What is the optimal load resistance that needs to be connect to the terminals of the PV panel?
b) With a schematic diagram, describe the working of a basic photovoltaic system for power generation. [5+5]

6.a) Explain how the energy produced by a wind turbine can be stored for re-use. What are the arrangements used for starting a Darrieus wind turbine?
b) Explain the working of vertical axis wind mill with a neat sketch. [5+5]

OR

7.a) Derive an expression for power extracted from wind.
b) A horizontal axis wind turbine is installed at a location having free wind velocity of 15 m/s. The 80 m diameter rotor has three blades attached to the hub. Find the rotational speed of the turbine for optimal energy extraction? [5+5]

8.a) Discuss the significant factors effecting of aerobic digestion and anaerobic digestion.
b) Classify wet and dry process of energy conversion technologies. [5+5]

OR

9.a) Write a short note on:
i) Geo-pressure resources
ii) Magma resources.
b) Enumerate the Prime movers used in Geo thermal power stations. [5+5]

10.a) Discuss the scope of utilizing ocean wave energy to generate electricity.
b) What are the various methods of tidal energy generation? Explain in detail. [5+5]

OR

11. Explain the advantages of fuel cell power sources. Draw a simple sketch of H_2-O_2, fuel cell and explain its working. [10]

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